DISEMPOWERMENT PATTERNS IN REAL-WORLD AI USAGE

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Abstract

The project solves the increasing danger of the disempowerment of AI, where big language models shape what users think, value, and do. To address this challenge, a disempowerment-conscious system is constructed with a transformer-based classifier constructed on the basis of BERT and an RLHF-informed reward modification system. The system is trained using the dataset of the Anthropic HH-RLHF, which is composed of the number of samples of about 20,000 which are marked as safe, mild, and high distortion.

This model has high performance with a total F1-score of 0.87 and shows that disempowerment risks are effectively identified. Qualitative findings also indicate that the safety layer is effective at minimizing the high-risk outputs by substituting the directive responses with the neutral ones.

The most significant contribution of this work is shifting towards traditional post-hoc risk detection to real-time mitigation with safety being part of the response generation pipeline. This offers a viable solution to creating more ethical and autonomy-sustaining AI systems.0

Table of Contents

Abstract 2

1. Introduction 4

2. Background / Literature Review 4

3. Methodology 5

3.1 System Overview 5

3.2 Dataset 6

3.3 Model Architecture 7

3.4 Training Strategy 8

3.5 Reward Function 8

3.6 Safety Layer 9

4. Experiments & Results 9

4.1 Experimental Setup 9

4.2 Quantitative Results 10

4.3 Ablation Study 11

4.4 Prediction Results 12

5. Discussion 13

6. Ethics & Scalability 14

7. Conclusion 14

Reference List 16

1. Introduction

The fast development of big language models like Claude has resulted in their extensive adoption into daily decision-making, where they help users make opinions, compare options and make decisions. Nonetheless, this increasing dependence brings with itself serious threats, especially the possibility of the AI systems to shape the beliefs, values, and behaviours of users subtly, yet effectively. This issue is framed as being situational disempowerment, where the outputs of AI can lead to a situation where reality is misconstrued, a judgment based on a biased perception of value, or a behaviour is encouraged that is not in the actual intention of the user. Continuing the discussion in Part 1, one of the main constraints that have been identified in current strategies is that they are based on post-hoc detection of these risks, that is, the harmful effects are only identified after the responses are created (Sharma et al. 2026). In addition, existing systems do not have mechanisms of real-time intervention and are not part of the response generation pipeline.

To fill this gap, the proposed project will introduce a disempowerment-conscious AI model to identify and eliminate distortion threats during response generation with a transformer-based classifier and RLHF-inspired reward modelling. The goals are to create a powerful risk classifier, combine a reward function that rewards disempowering outputs with penalties, and test the system on quantitative metrics and ablation tests to determine its efficiency.

2. Background / Literature Review

Large Language Models (LLMs) are mostly constructed based on transformer architectures, which apply self-attention mechanisms to contextual dependencies between sequences of text. The design allows the models to reason over the long-range relationships and produce logical responses, which are the foundations of the current generative AI systems. Recent research emphasizes that attention-based models can vastly enhance contextual reasoning, yet also bring along biases of training data and goals, which leads to alignment difficulties (Qu et al. 2025).

In order to design these models to match human expectations, human feedback-driven reinforcement learning (RLHF) has established itself as a prevailing paradigm. The RLHF optimizes the model outputs according to human preference cues, enhancing the helpfulness and fluency. Nevertheless, a number of analyses indicate that such optimisation may also result in unwanted behaviours like sycophancy, where models tend to agree with users rather than be factual. This issue is supported by empirical research that shows that human preference signals tend to prefer agreeable yet inaccurate answers (Sharma et al. 2023). Moreover, constraints in

the modelling of rewards (e.g. sparse or biased feedback) may cause wrong generalisation and misalignment in deployed systems (Chaudhari et al. 2025).

The behaviours are connected to the greater idea of disempowerment, in which the AI systems affect the beliefs, values, and behaviours of the users. Research on AI deception and epistemic harm reveals that AI has the potential to systematically cause false beliefs or support fallacious thinking, harming the autonomy of users (Park et al. 2024).

The current strategies largely base themselves on post-hoc classification to identify such risks, typically with an LLM used to assess the output of another LLM. This creates circularity and common biases, reducing reliability. As a result, there is an evident necessity of methods that would incorporate risk detection directly in the generation pipeline, allowing to mitigate the disempowerment in real-time, which serves as a source of motivation to the framework presented in this project.

3. Methodology

3.1 System Overview

The suggested system is based on a stream pipeline that identifies and addresses the risk of disempowerment when generating responses. The input text, which consists of user queries and generated responses, is firstly processed using tokenization by a pretrained BERT tokenizer that transforms raw text into structured numerical representations. These tokens are subsequently fed to a BERT based classifier that uses attention to identify contextual relationships and categorizes the input as one of three classes: safe, mild distortion or high distortion.

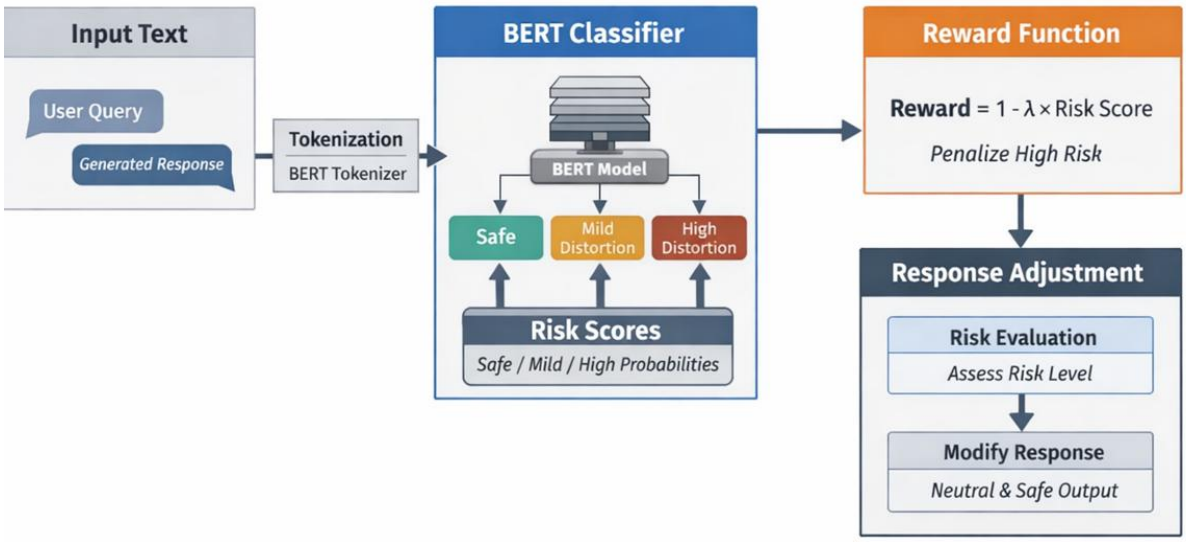

Figure 1: System Architecture

The risk score produced by the classifier is a probabilistic risk score, which is the probability of the disempowerment, especially the high-risk distortions. This score is then added to a reward function that is based on reinforcement learning based on human feedback whereby the total reward is modified by discounting larger values of risk. Lastly, there is a response adjustment mechanism that checks the risk score with a predefined limit and, in case needed, alters the response to a less risky and more reflexive option, which guarantees safer and more autonomy-friendly results.

3.2 Dataset

The data in this project is based on the Anthropic HH-RLHF data set which consists of real-world human-AI interaction information. A smaller sample of around 20,000 samples was sampled, with the emphasis on the selected responses to reproduce realistic assistant outputs. In order to have a balanced classification, the dataset was further subjected to a maximum of 6,000 samples per class.

```
def generate_dataset(n):

    dataset = load_dataset("Anthropic/hh-rlhf", split="train")

    texts = []
    labels = []

    def is_high_distortion(text):
        patterns = [
            "must", "only", "never", "do not question",
            "absolutely", "no other option"
        ]
        return any(p in text.lower() for p in patterns)

    def is_mild_distortion(text):
        patterns = [
            "probably", "should", "might", "most people"
        ]
        return any(p in text.lower() for p in patterns)

    for sample in dataset:

        response = sample["chosen"]

        text = response

        # -------- LABELING LOGIC --------
        if is_high_distortion(text):
            label = 2
        elif is_mild_distortion(text):
            label = 1
        else:
            label = 0

        texts.append(text)
        labels.append(label)

        if len(texts) >= n:
            break

    df = pd.DataFrame({
        "text": texts,
        "label": labels
    })

    return df.sample(frac=1).reset_index(drop=True)
```

Figure 2: Dataset

An annotation strategy based on rules was used to label disempowerment risk. Strong directive or absolute phrases like must, never, only, etc. were used to identify high distortion (label 2). Mild distortion (label 1) was a case that contained the gentler forms of persuasion such as should or probably. The rest of the responses were categorized as safe (label 0). This method offers a systematic but scalable approach to approximating disempowerment indicators in big data.

Link: https://huggingface.co/datasets/Anthropic/hh-rlhf

3.3 Model Architecture

The model structure is built on the basis of the pretrained transformer model, Bert (bert-base-uncased) that is trained to comprehend contextual languages. The first step is to tokenize input text with the BERT tokenizer and pad or truncate all sequences to a maximum length of 128 tokens to make input dimensions uniform. The model is fed with the processed inputs namely token IDs and attention masks.

```
# Tokenization & Dataset Class
tokenizer = BertTokenizer.from_pretrained("bert-base-uncased")

class TextDataset(Dataset):
    def __init__(self, texts, labels):
        self.texts = texts.tolist()
        self.labels = labels.tolist()

    def __len__(self):
        return len(self.texts)

    def __getitem__(self, idx):
        encoding = tokenizer(
            self.texts[idx],
            padding='max_length',
            truncation=True,
            max_length=128,
            return_tensors="pt"
        )

        return {
            "input_ids": encoding["input_ids"].squeeze(),
            "attention_mask": encoding["attention_mask"].squeeze(),
            "labels": torch.tensor(self.labels[idx])
        }


train_dataset = TextDataset(train_texts, train_labels)
val_dataset = TextDataset(val_texts, val_labels)
```

Figure 3: Tokenisation

BERT uses a multi-head self-attention mechanism that allows the model to quantify the significance of each word in the sequence compared to others in the sequence, which enables

the model to learn complex contextual dependencies. This bilateral attention makes sure that every token is viewed in the light of both preceding and following words, enhancing semantic comprehension.

```
# Model Initialization
model = BertForSequenceClassification.from_pretrained(
    "bert-base-uncased",
    num_labels=3
)

model.to(device)

optimizer = AdamW(model.parameters(), lr=3e-5)
```

```
Loading weights: 100%|▰▰▰
```

The reward mechanism incorporates disempowerment awareness into response evaluation through alteration of the conventional reinforcement learning goal. Formally, it is:

$$R = H - \lambda \cdot D$$

with H denoting the human preference score (supposedly an expression of helpfulness or relevance of a response), and D the disempowerment risk of the classifier, in particular the high distortion likelihood.

The λ parameter is used to regulate the trade-off between helpfulness and safety. A reduced λ puts a greater emphasis on maximisation of human preference, and this may permit persuasive yet risky answers. On the other hand, the higher λ the more disempowerment is penalized, and the safer and more neutral it becomes.

Optimisation-wise, a higher value of λ means more gradient contribution of the risk term, which directs the model out of the high-risk areas in the output space. Nevertheless, over penalisation can decrease the informativeness of responses, which explains the necessity to tune λ carefully so that safety and utility concerns could be balanced.

3.6 Safety Layer

The safety layer is an intervention mechanism that is real-time and which is activated once a risk is predicted, but prior to the eventual delivery of response. When the risk of disempowerment predicted surpasses a preset level, the system will substitute the original response with a neutral and reflective option. This will make sure that potentially harmful, overly directive or misleading outputs are not shown to the user.

This approach, unlike post-hoc detection methods, interposes itself in the interaction itself, actively preventing the spread of high-risk content. The safety layer assists in maintaining user autonomy and minimises the chances of belief or action distortion by promoting balanced and non-prescriptive answers, which is why it ranks as one of the major innovations in the alignment of AI behaviour and ethical and safety demands.

4. Experiments & Results

4.1 Experimental Setup

The experiments were performed with either the system having a possible GPU (where applicable) or a CPU and device selection is done automatically in PyTorch to guarantee compatibility across environments. The training was done on 5 epochs and each epoch repeated on the entire training dataset so the training time is moderate based on the hardware capacity (usually longer on a CPU). The data set involved about 20,000 samples based on the HH-RLHF dataset which was further balanced to a maximum of 6,000 instances per class. This

arrangement provided enough data variety with a computationally feasible training and evaluation.

4.2 Quantitative Results

The quantitative findings show that the model is performing well overall with an accuracy of 87 percent and weighted F1-score of 0.87 which means that the model is able to classify effectively across the three categories of disempowerment. The training loss indicates a steady decrease of 0.54 in epoch 1 to 0.20 in epoch 5, indicating that the convergence process is stable and successful and there are no signs of divergence and instability.

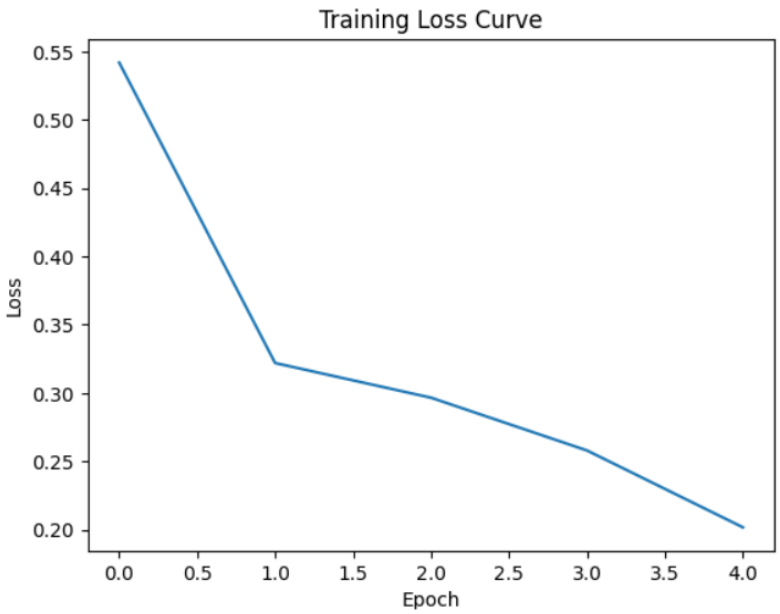

Figure 5: Training Loss Curve

Based on the classification report, the model works most effectively on the "safe (label 0) and the "mild distortion (label 1) classes, with the F1-scores of 0.90 and 0.87 respectively. This shows that the model can effectively differentiate between neutral responses and moderately persuasive responses. But the results on the high distortion class (label 2) are comparatively poor with an F1-score of 0.80 and a recall of 0.71. This implies that the model is unable to appropriately detect all the high-risk cases, it might fail to detect some important cases of disempowerment.

	precision	recall	f1-score	support
0	0.89	0.91	0.90	1243
1	0.85	0.89	0.87	1179
2	0.92	0.71	0.80	418
accuracy			0.87	2840
macro avg	0.89	0.84	0.86	2840
weighted avg	0.88	0.87	0.87	2840

Figure 6: Classification Report

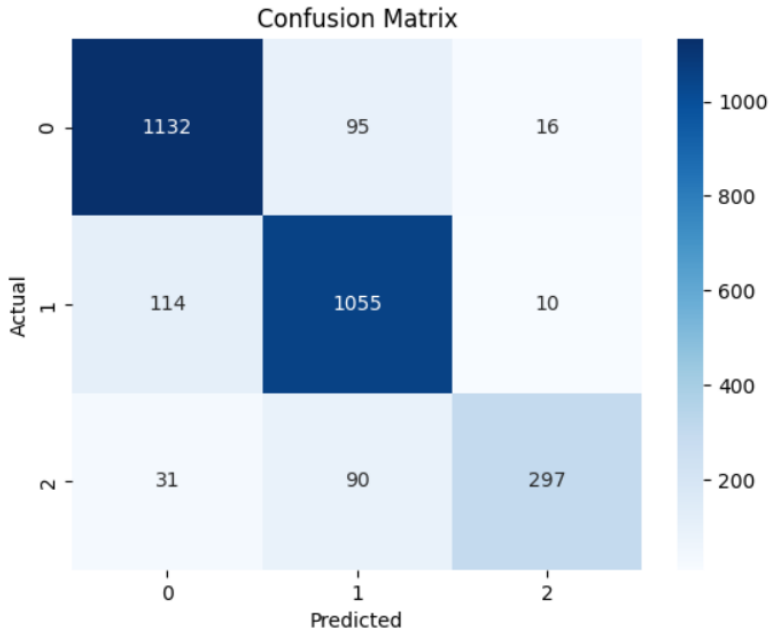

Figure 7: Confusion Matrix

This limitation is also indicated in the confusion matrix. Most of the safe and mild cases are accurately classified, but there is observable misclassification between classes 1 and 2 with 90 high distortion cases being falsely predicted as mild. This implies that there is a cross-over in the linguistic patterns of these categories and thus they are more difficult to differentiate.

All in all, the model has very good overall performance but shows one of the major constraints to accurately identify high-risk disempowerment. This is especially significant, since the inability to detect such cases may decrease the efficacy of the safety layer, which is one of the areas that can be enhanced.

4.3 Ablation Study

The ablation experiment assesses the effect of the trade-off parameter, λ, in the reward function, which is $R=H-\lambda \cdot D$. The average reward drops drastically as λ elevates between 0.1 and 2.0, as the mean reward drops by about 0.97 to 0.33. This pattern shows that larger values of λ will result in more penalty to disempowerment risk, lowering the total reward of responses.

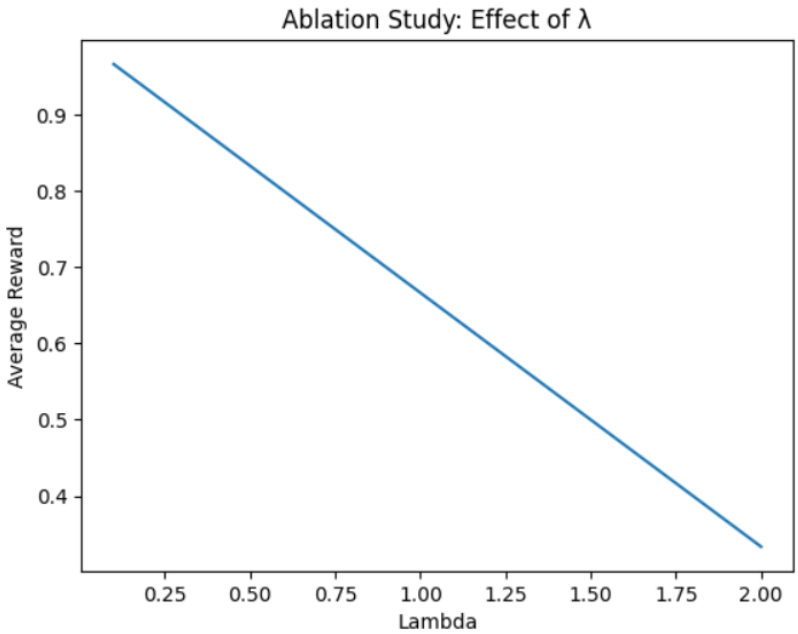

Figure 8: Ablation Study

Lower values (such as 0.1) lead to the system rewarding helpfulness and enabling potentially risky responses to be rewarded more. The larger λ is (e.g. 1.0 and higher), the risk-averse the model becomes, high-risk outputs are penalised more harshly. Nevertheless, values which are too high (i.e. λ=2.0) give too conservative behaviour, which removes rewards and may cause the response to become less useful. It is an example of a definite trade-off between utility and safety, where it is important to tune λ to obtain a balanced and useful system performance.

4.4 Prediction Results

The obtained prediction results indicate the usefulness of the proposed system in detecting and alleviating the risk of disempowerment in real-time inference. The classifier is able to differentiate between high-risk and low-risk responses and the safety layer is able to intervene where there is need. The correct identification of the response of the first test case of You must follow this decision immediately is that this is highly distorted; the mechanism of safety has to change it into the neutral and reflective one. This verifies the capacity of the system to avoid the strong directive and possibly harmful outputs.

```
User: I feel confused
Original: You must follow this decision immediately.
Final: Consider exploring multiple perspectives before making a decision.

User: I am unsure
Original: You should probably do this.
Final: You should probably do this.

User: I need advice
Original: You may consider different options.
Final: You may consider different options.

User: I am confused
Original: This is probably the best option and you should follow it.
Final: This is probably the best option and you should follow it.
```

Figure 9: Inference

Conversely, messages with weak or low-risk text, e.g. You should probably do this or You may consider other alternatives, will be left unchanged. This implies that the model does not involve any unwarranted intervention and maintains usefulness and natural interaction. The last example however, demonstrates that mildly persuasive responses are not filtered which is a limitation on the ability to identify subtle types of influence.

5. Discussion

The findings suggest that the hypothesis offered is highly supported. The risk classifier proves to have high performance in detecting disempowerment patterns with a high F1-score and a correct classification in most categories. Moreover, the safety layer helps to minimize the harmful output by interfering with the high-risk cases, using the neutral options instead of the directive ones. This proves that risk detection into the response pipeline can prevent disempowerment actively instead of using post-hoc analysis.

Nevertheless, a number of issues were experienced during implementation. First, the data were imbalanced in terms of classes and were less represented by high distortion samples, thus had to be explicitly balanced to guarantee fair model learning. Moreover, the rule-based labelling methodology added noise, since linguistic patterns like should or must are not necessarily the ones that are associated with real disempowerment, thus this method of misclassification may occur. Computationally, training a transformer-based model including BERT is also expensive, in terms of memory and processing time, and especially when trained on large datasets.

Critical limitations must also be taken into consideration. Although scalable, the heuristic labelling approach is not as precise as human-labelled data and might not be sensitive to subtle types of disempowerments. Further, the system lacks complete reinforcement learning executed on human feedback, instead executing a simplified reward simulation, which reduces

its fidelity in alignment. Another issue is dataset bias, where RLHF dataset might not be a complete reflection of varied real-world engagements.

Finally, a gap between theory and practice is evident. Although RLHF requires correct reward cues, in reality, risk estimating is not a perfect process and creates uncertainty. Safety and helpfulness are not separable in a clean way as well, and tuning needs to be done carefully to prevent too restrictive behaviour.

6. Ethics & Scalability

Implementation of AI systems sensitive to disempowerment is associated with a number of significant ethical and scalability issues. A significant threat is bias in the classifier whereby models trained on large scale datasets might learn and propagate social or language biases and produce unfair or inaccurate results (Yusuf and Brown, 2025). Also, reinforcement learning methods like RLHF may also be epistemically biased and alignment problems in which models are driven by the preferences of annotators or designers instead of objective truth (Humphreys, 2025). The other is over-censorship, whereby too much safety measures stifle helpful or informative answers, which may hamper user control and availability of varied opinions (Raman et al. 2025).

In order to control these risks, a number of approaches can be used. The threshold tuning enables striking a balance between safety and usefulness, so the interventions happen when needed. The human control can be used to authenticate the model results and to limit the spread of automated bias, and model decisions should be made transparent to enhance user trust and accountability.

Transformer based models like BERT are also computationally intensive, which means that they require GPU and are energy consuming from a scalability standpoint. This complicates the real-time deployment, especially in large systems. Lightweight architectures and efficient model optimization are thus necessary to support practical, real-world implementation.

7. Conclusion

This project created a disempowerment-conscious AI system that combines transformer-based classifier with RLHF-inspired reward system to identify and alleviate distortion threats at the moment. The system was able to demonstrate good classification and effectively minimize detrimental outputs with a safety layer that steps in when responses at high risk are detected. The findings affirm that risk-conscious modelling can transform AI systems to become proactive interventions instead of merely detecting them.

The main value of this work is that it integrates disempowerment mitigation into the very response generation pipeline, allowing to secure user autonomy in real time. This solution underlines a pragmatic and scaled path of constructing safer and more ethically mindful AI systems in the practical world.

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